FIRST-ORDER REACTION KINETICS

Data Analysis and Mathematical Modeling

Chemical Engineering Portfolio Project

Prepared by: Zhe Wang
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Abstract

This project analyzes simulated concentration-versus-time data for a first-order chemical reaction. The concentration data were transformed using the natural logarithm and analyzed using linear regression. The regression produced a slope of -0.09888 min^{-1}, giving an estimated reaction-rate constant of 0.09888 min^{-1}. The resulting R^2 value was 0.99967, indicating a strong linear relationship in the transformed data. An exponential model was then developed to represent reactant concentration as a function of time. The project demonstrates mathematical modeling, data analysis, regression, and chemical-kinetics concepts.

1. Introduction

Chemical reactions are an important part of chemical engineering, and understanding how the concentration of a reactant changes over time is essential for analyzing and designing chemical processes. One common reaction model is first-order kinetics, in which the reaction rate is directly proportional to the concentration of the reactant. This relationship can be represented using an exponential decay model.

In this project, simulated concentration-versus-time data for a first-order reaction were analyzed using Google Sheets. The concentration data were transformed using the natural logarithm and analyzed through linear regression to determine the reaction-rate constant. The resulting mathematical model was then compared with the concentration data to evaluate how well the first-order model represented the reaction behavior.

2. Theoretical Background

For a first-order reaction:

$$\mathbf{A \rightarrow B}$$

the integrated first-order rate law is:

$$\mathbf{C_a(t) = C_{a,_0}\ e\text{^}(-kt)}$$

where $C_a(t)$ is the concentration of reactant A at time t, $C_{a,_0}$ is the initial concentration, k is the reaction-rate constant, and t is time.

Taking the natural logarithm gives the linear form:

$$\mathbf{\ln(C_a) = \ln(C_{a,_0}) - kt}$$

This has the form y = mx + b. Therefore, the slope of a plot of $\ln(C_a)$ versus time is equal to −k.

3. Methodology

A simulated dataset containing reactant concentrations at two-minute intervals from 0 to 20 minutes was analyzed. The dataset was used for mathematical modeling and was not collected from a physical laboratory experiment. The data were entered into Google Sheets, graphed, transformed using ln(concentration), and analyzed with linear regression.

Time (min)	Concentration (mol/L)	ln(Concentration)
0	1.000	0.0000
2	0.813	-0.2070
4	0.676	-0.3916
6	0.541	-0.6143

8	0.451	-0.7963
10	0.373	-0.9862
12	0.306	-1.1842
14	0.248	-1.3943
16	0.208	-1.5702
18	0.164	-1.8079
20	0.139	-1.9733

4. Results

The concentration decreased from 1.000 mol/L at 0 minutes to 0.139 mol/L at 20 minutes, showing the expected decrease in reactant concentration.

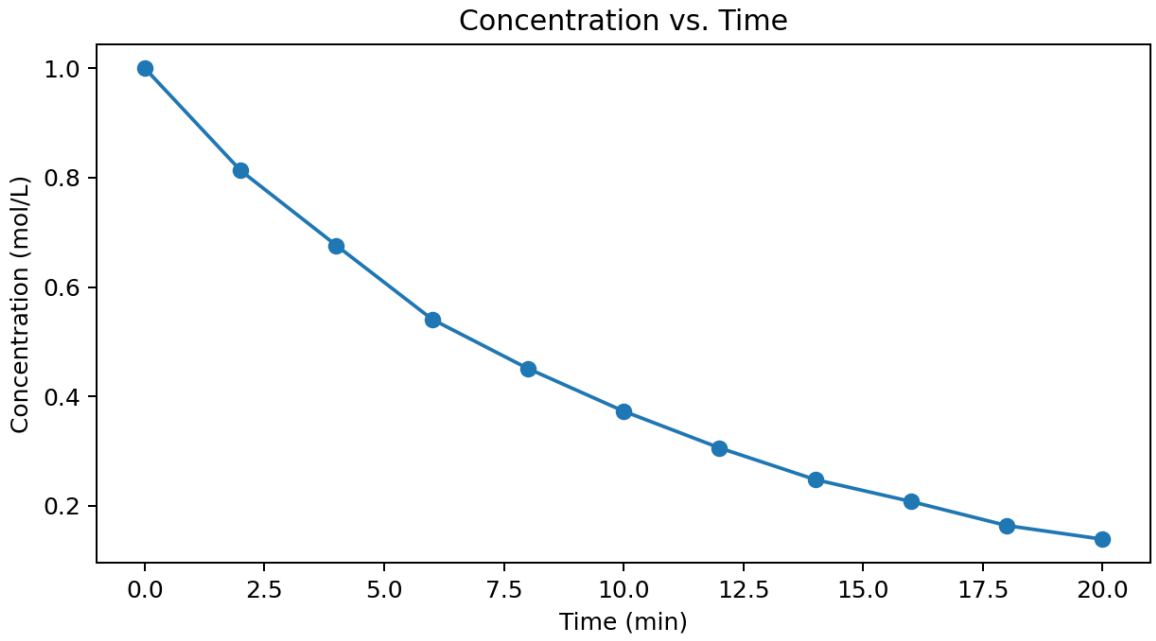


Figure 1. Concentration decreases with time.

Linear regression of ln(concentration) versus time produced:

$$\mathbf{\ln(C_a) = -0.09888t - 0.0044}$$

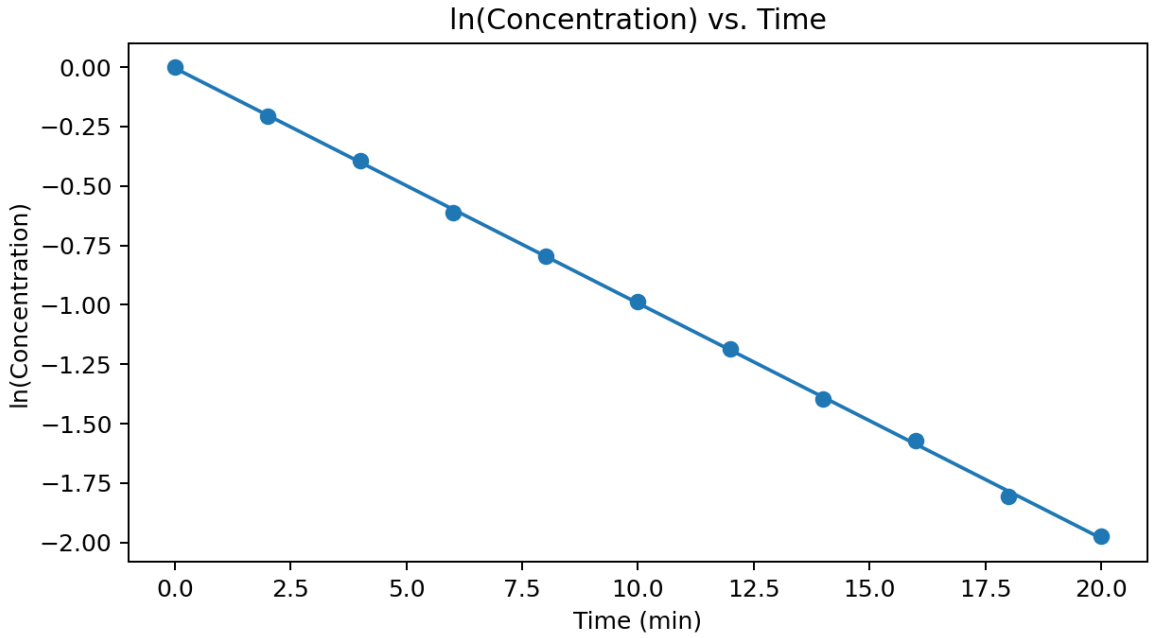


Figure 2. Linearized first-order reaction model.

Because the slope is −k, the estimated reaction-rate constant is k = 0.09888 min^{-1}. The fitted initial concentration is approximately 0.9956 mol/L. The regression R^2 value is 0.99967.

5. Mathematical Model

$C_a(t) = 0.9956e^{(-0.09888t)}$

This equation can be used to estimate the reactant concentration as a function of time within the range represented by the dataset.

6. Discussion

The approximately linear ln(concentration)-versus-time relationship supports the use of a first-order kinetic model for this dataset. The negative slope indicates that the reactant concentration decreases with time, while the magnitude of the slope gives the reaction-rate constant. The R^2 value of 0.99967 indicates that the transformed data closely follow the fitted linear relationship.

The model and data are not identical at every point, which is expected when analyzing data containing variation. Also, a high R^2 value describes the quality of the regression fit; it should not be interpreted as a direct measure of experimental accuracy.

7. Limitations

The main limitation is that the dataset was simulated rather than collected from a physical laboratory experiment. Therefore, the project demonstrates data analysis and mathematical modeling rather than experimental validation. A laboratory study could introduce additional sources of variation such as measurement uncertainty, temperature changes, mixing conditions, and instrument limitations.

8. Conclusion

This project demonstrated how concentration data can be analyzed to evaluate first-order reaction behavior. Linear regression of the logarithmically transformed data produced a reaction-rate constant of 0.09888 min^{-1}. The resulting exponential model, $C_a(t) = 0.9956e^{(-0.09888t)}$, closely represented the

simulated dataset, with an R^2 value of 0.99967. The project provided experience with data organization, mathematical modeling, linear regression, graphing, and interpretation of chemical-engineering data.

9. Portfolio Classification

This project should be presented as a data-modeling and mathematical-analysis project, not as a laboratory experiment. Recommended portfolio title:

First-Order Reaction Kinetics — Data Analysis and Modeling